

OP-förstärkaren

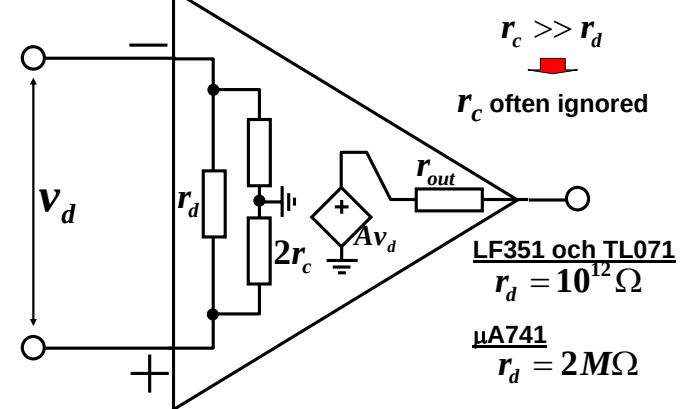
Icke Ideal OP

Viktor Öwall



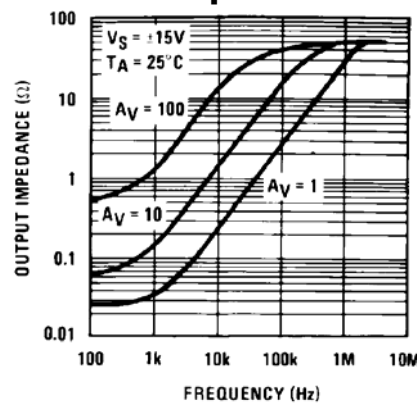
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Impedanser



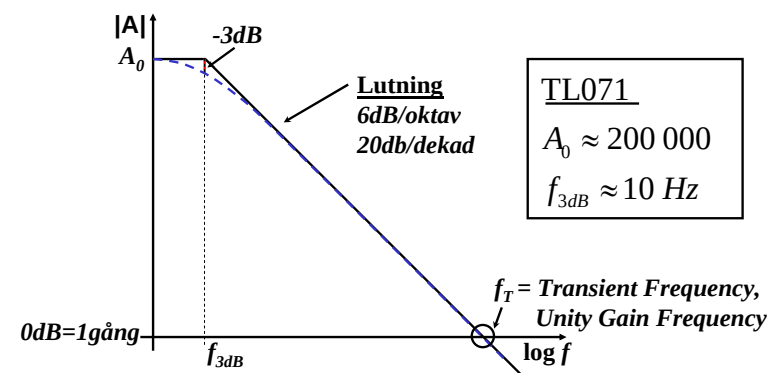
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Utimpedans



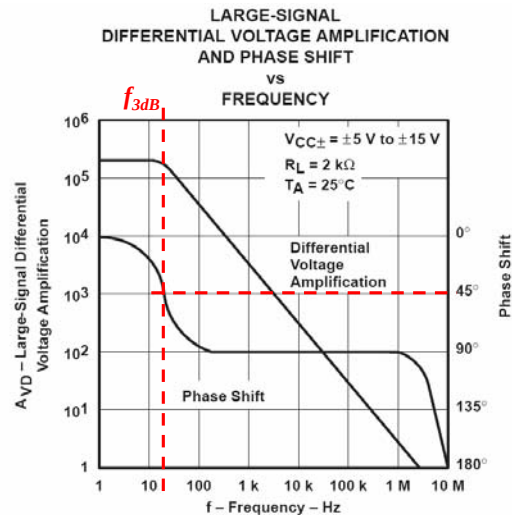
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OP:ns frekvensgång



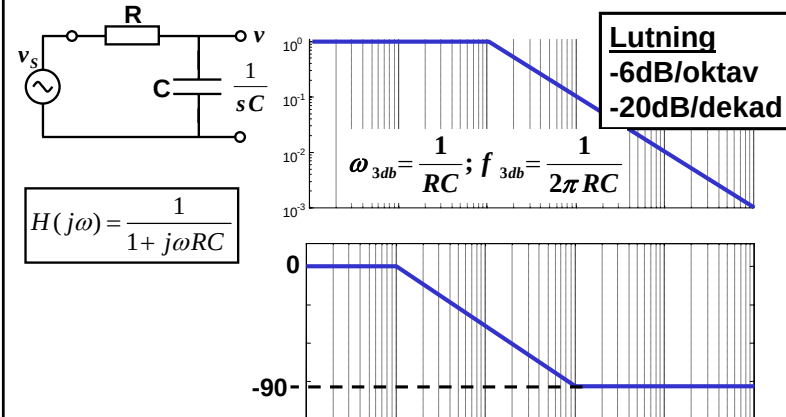
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OP: TL071



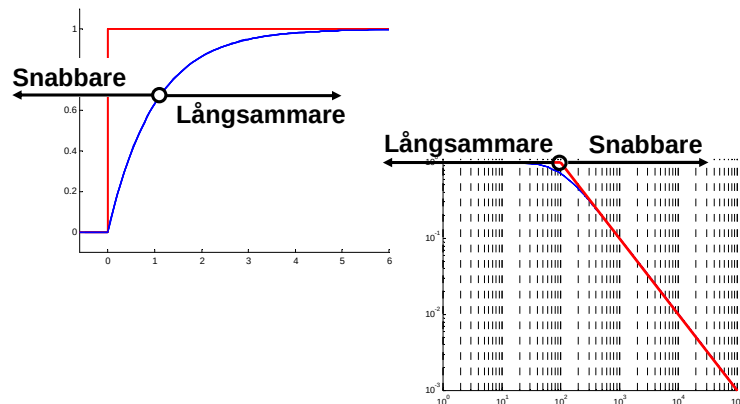
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Jämför lågpäss RC-länk



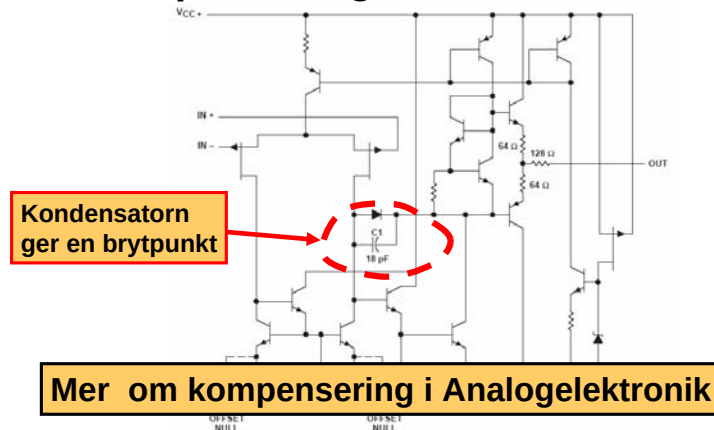
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Jämförelse Tid - Frekvens



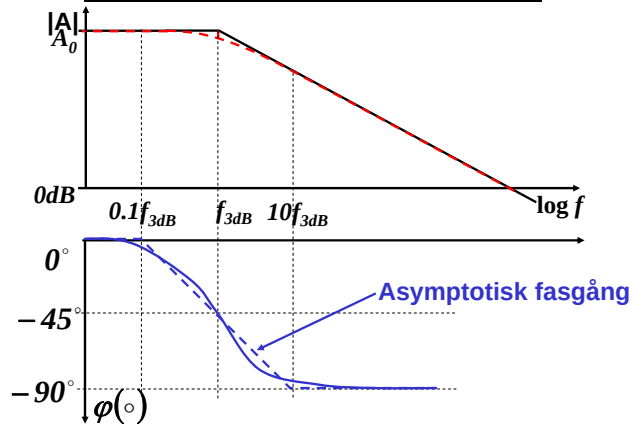
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Kompensering av OP TL071



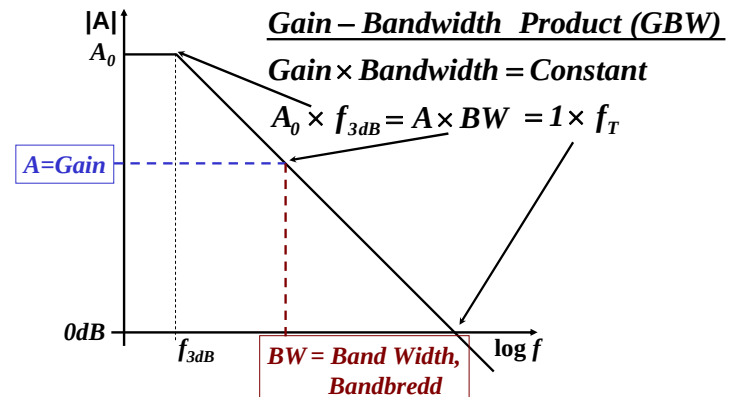
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OP:ns frekvensgång med fas

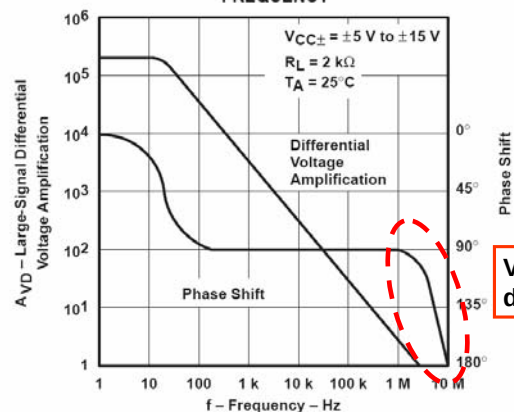


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Förstärkning-Bandbredd



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LARGE-SIGNAL
DIFFERENTIAL VOLTAGE AMPLIFICATION
AND PHASE SHIFT
vs
FREQUENCY

Vad säger detta oss?

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Bode-plot – OP741

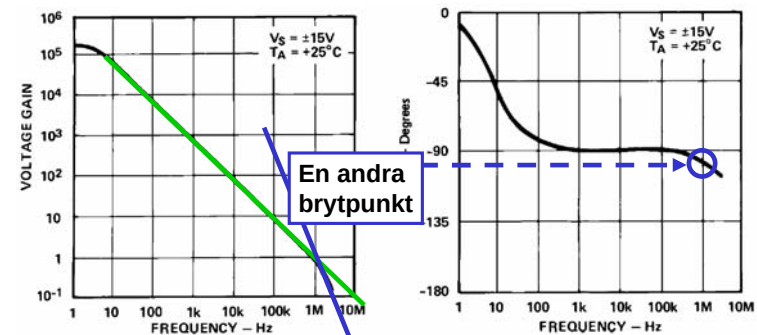
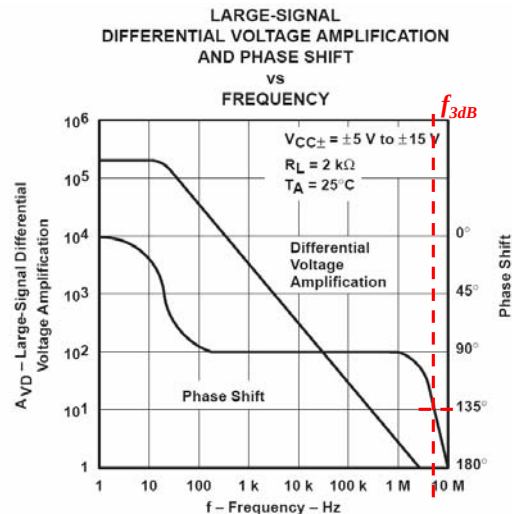


Figure 3. Open-Loop Gain vs. Frequency Figure 4. Open-Loop Phase Response vs. Frequency

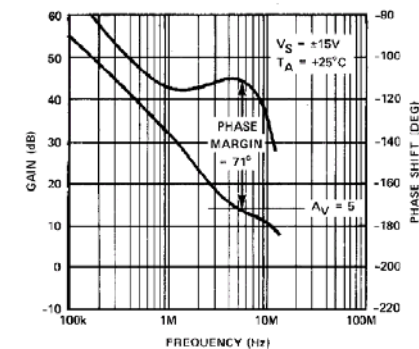
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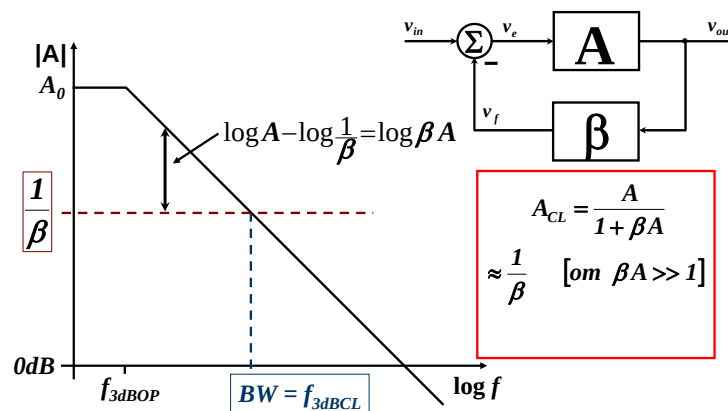
Bode-diagram OP37

**GAIN, PHASE SHIFT vs
FREQUENCY**



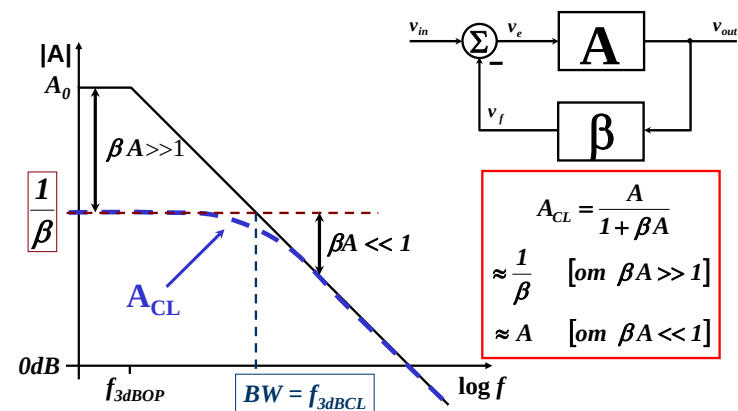
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Förstärkning-Bandbredd, forts.

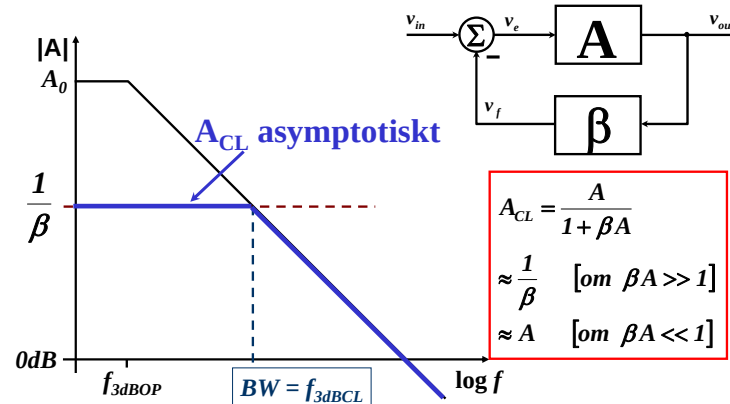


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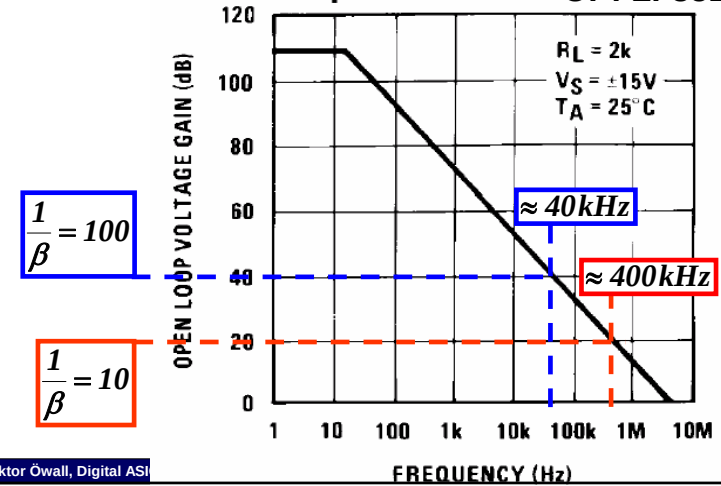
Förstärkning-Bandbredd, forts.



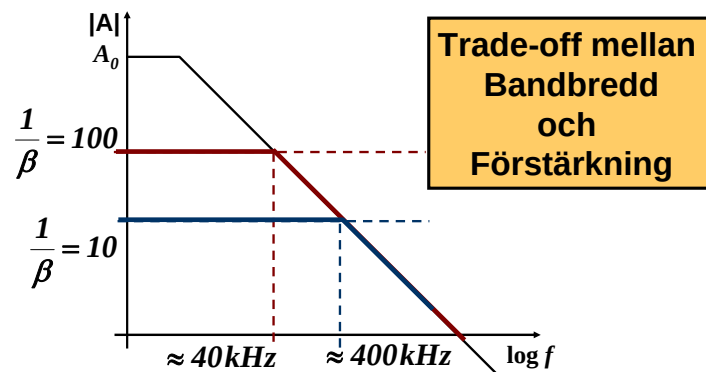
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Förstärkning-Bandbredd, forts.

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Open Loop Frequency Response
OP: LF351

Viktor Öwall, Digital ASIC

Förstärkning-Bandbredd, forts.

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Laplace och Bode**RC-nät**

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Överföringsfunktion – $H(s)$

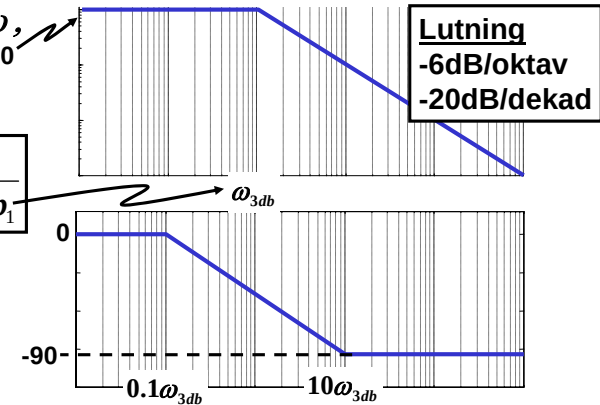
$$H(s) = \underset{\text{Förstärkningsfaktor}}{V} \frac{\overset{\text{Nollställen}}{(s - n_1)(s - n_2)(s - n_3) \dots (s - n_M)}}{\underset{\text{Poler}}{(s - p_1)(s - p_2)(s - p_3) \dots (s - p_N)}}$$

Pol i Bodediagram

Byt s mot $j\omega$,
nivå när $\omega \rightarrow 0$

$$H(s) = \frac{1}{s - p_1}$$

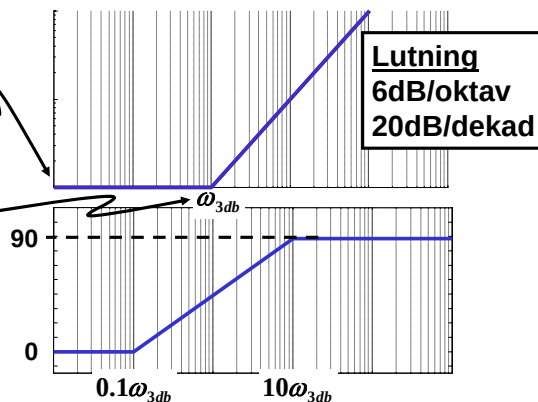
$$\omega = 2\pi f$$



Nollställe i Bodediagram

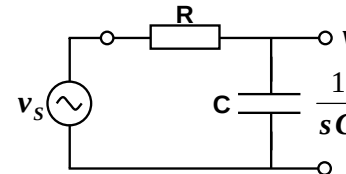
Byt s mot $j\omega$,
nivå när $\omega \rightarrow 0$

$$H(s) = s - n_1$$



Lågpas-Integrerande nät

Spänningsdelning

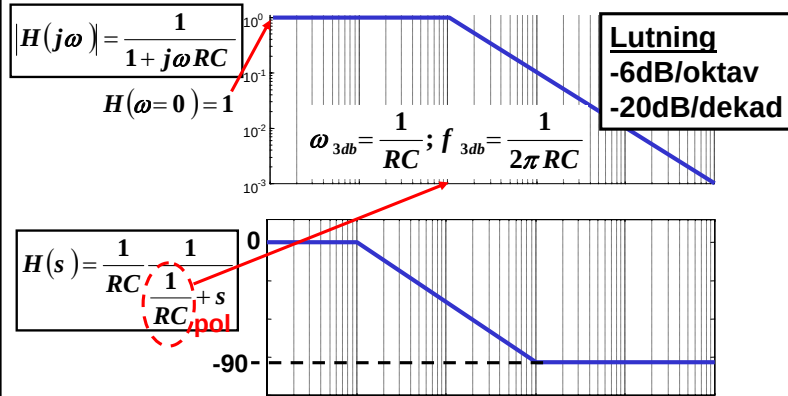


$$v = \frac{1/sC}{R + 1/sC} v_s$$

$$v = \frac{1}{1 + sRC} v_s$$

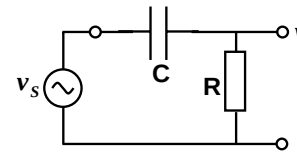
$$H(s) = \frac{1}{1 + sRC} = \frac{1}{RC} \frac{1}{s + \frac{1}{RC}}$$

LP - Bodediagram



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Pol och Nollställe - HP



$$H(s) = \frac{s RC}{1 + s RC}$$

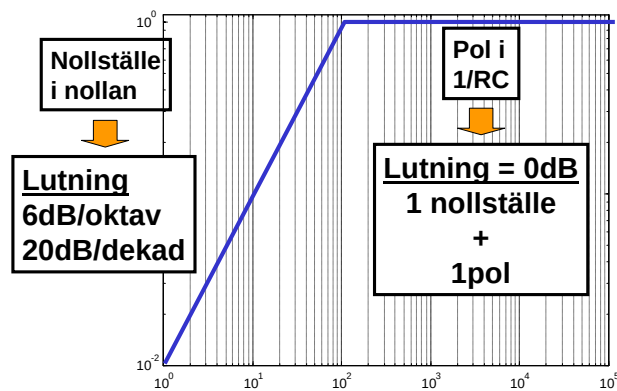
Nollställe i nollan

$$H(s) = \frac{s}{s + \frac{1}{RC}}$$

Pol

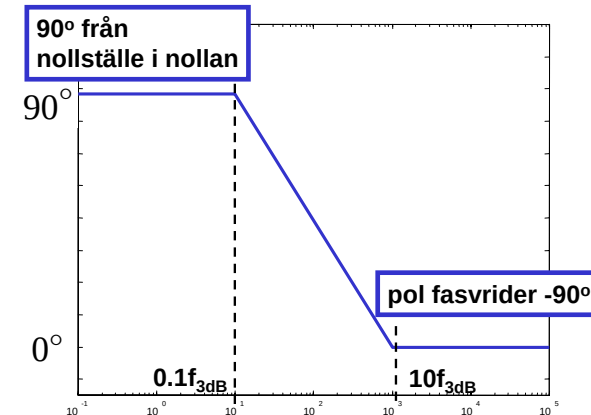
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Bodediagram - HP



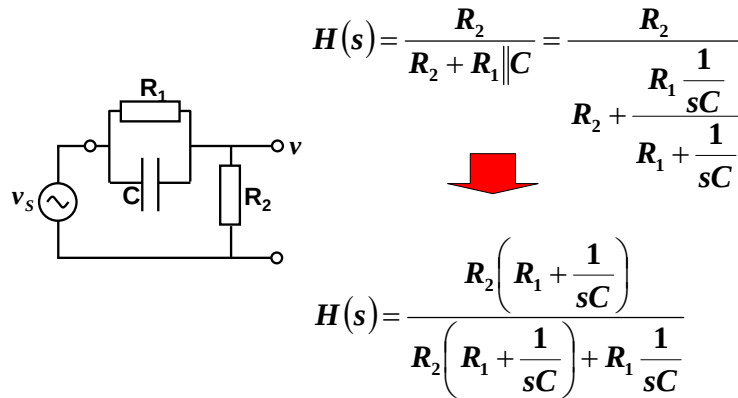
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Bodediagram- HP

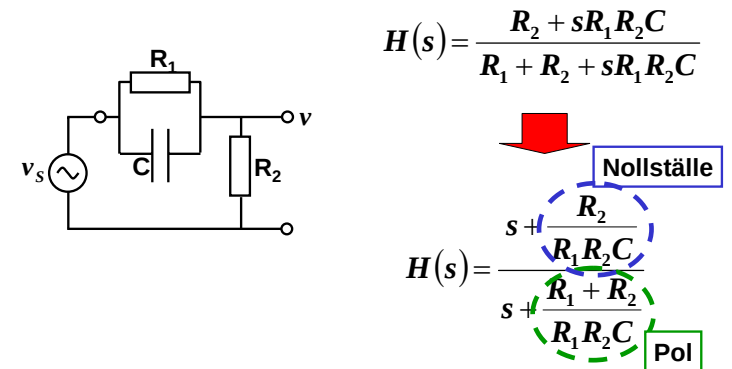


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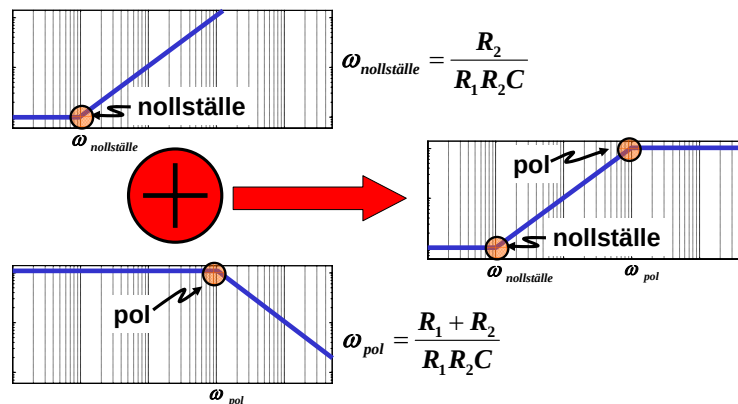
Pol och Nollställe - RCR



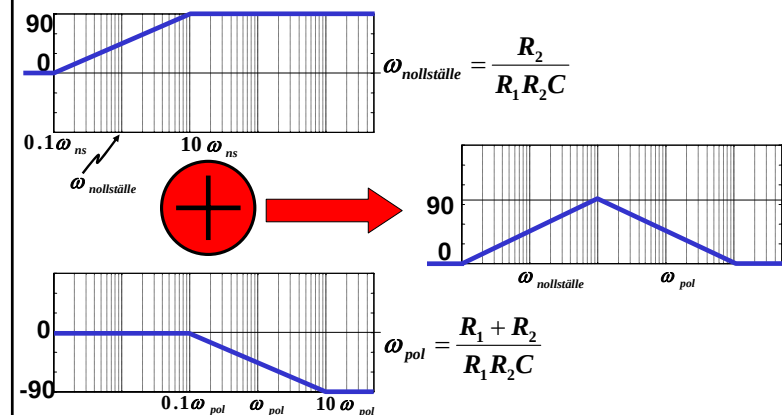
Pol och Nollställe - RCR



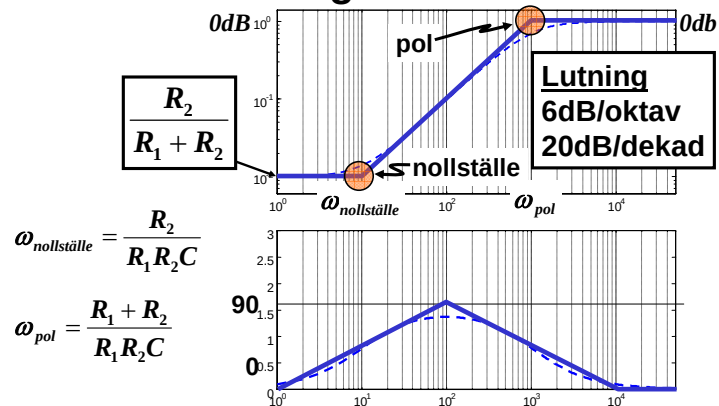
Amplitud: Nollställe + Pol



Fas: Nollställe + Pol



Bodediagram RCR-nät



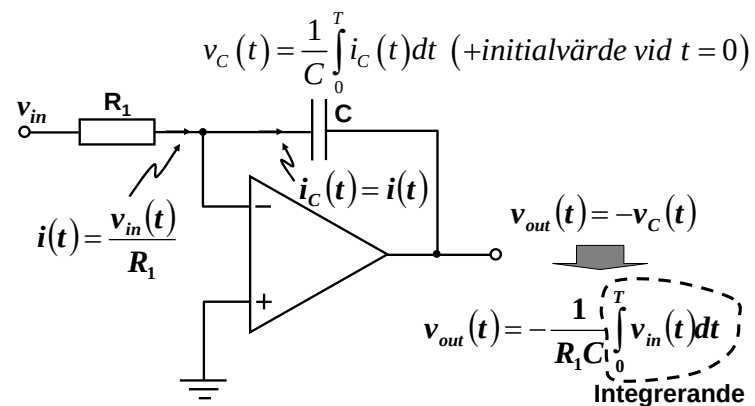
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Integratorn och Derivatorn



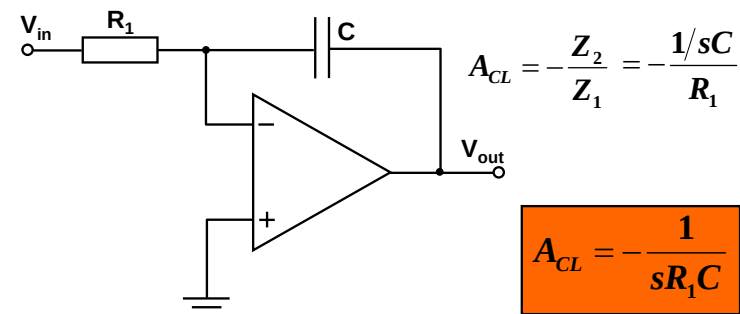
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Integratorn

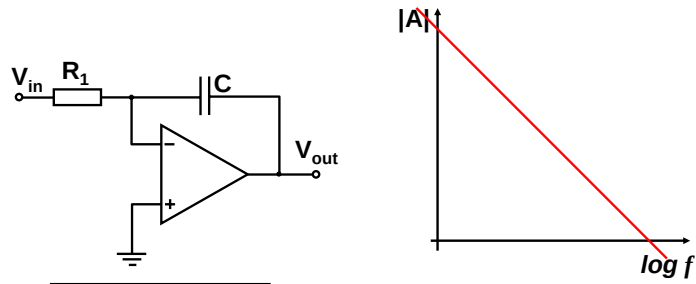


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Integratorn



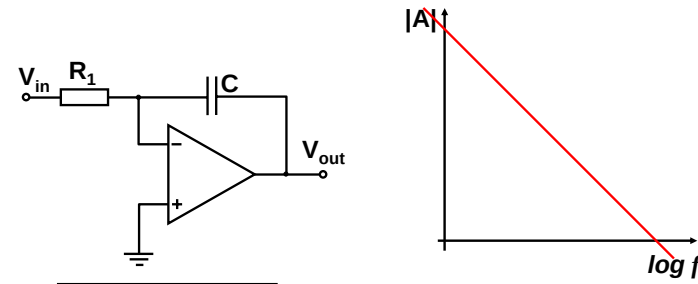
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Integratorn

$$A_{CL} = -\frac{1}{sR_1C}$$

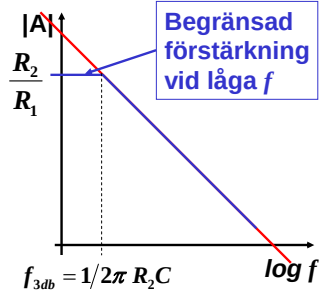
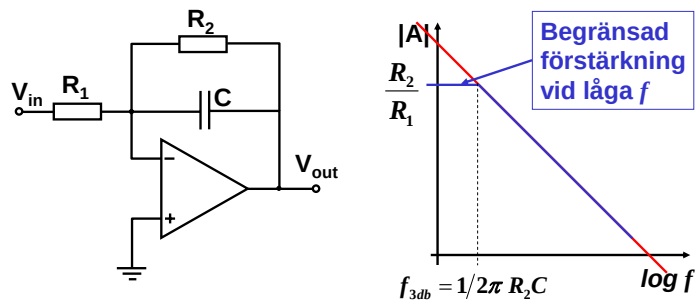
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2 minuters övning: Kan vi genom att studera Bode-plotten se några problem med denna koppling?



$$A_{CL} = -\frac{1}{sR_1C}$$

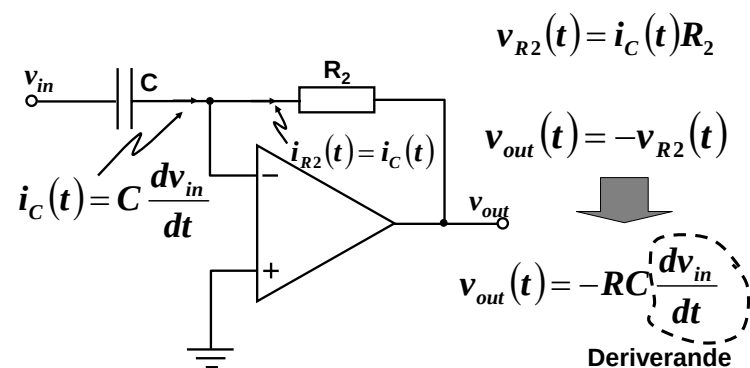
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AC-Integratorn

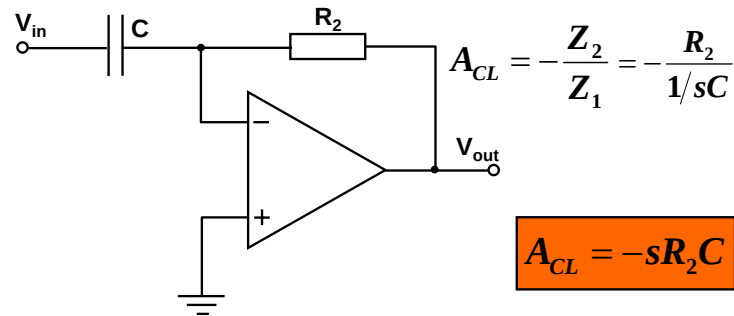
$$A_{CL} = -\frac{Z_2}{Z_1} = -\frac{R_2 \parallel C}{R_1} = -\frac{R_2}{R_1} \cdot \frac{1}{1 + sR_2C}$$

Lågpäss!

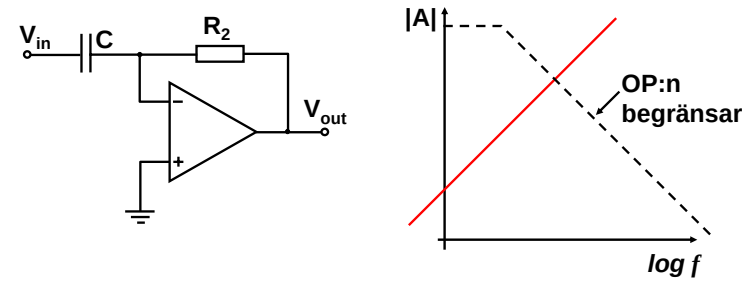
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Derivatoren-Differentiatorn

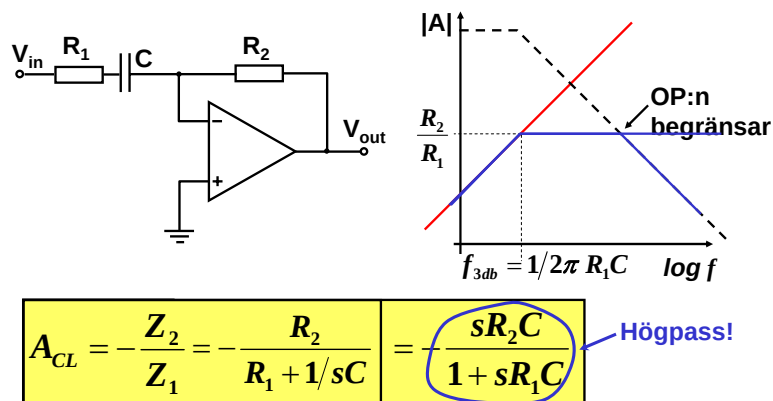
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Derivatorn-Differentiatorn

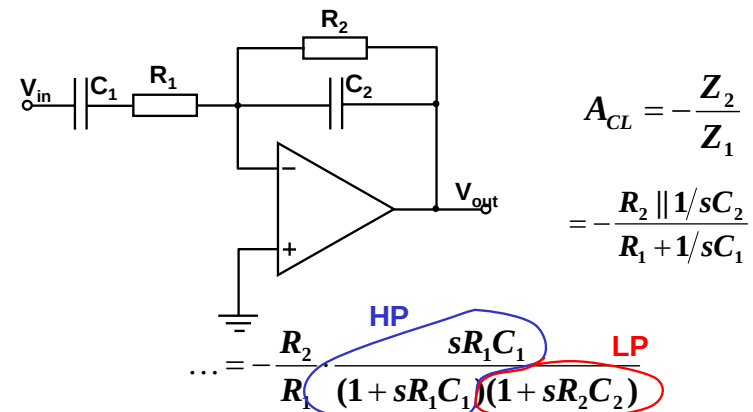
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Derivatorn-Differentiatorn

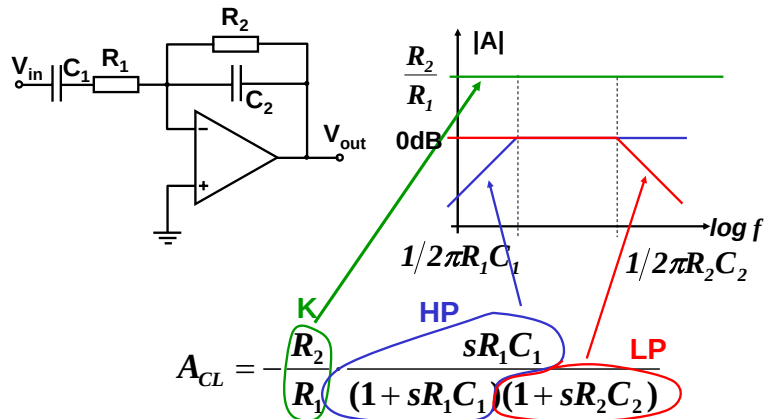
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Low-frequency Differentiator

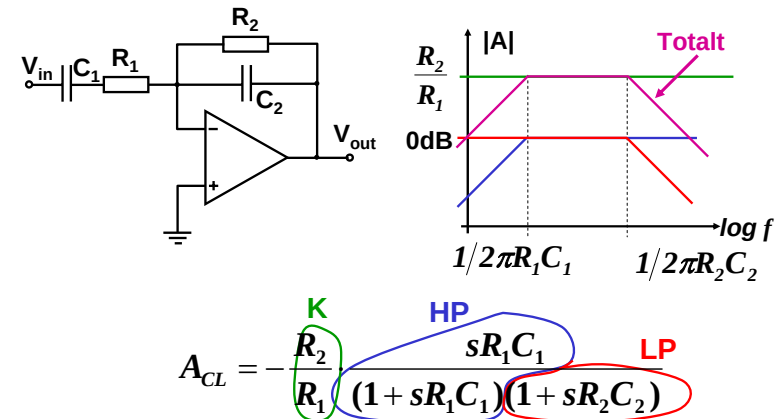
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Bandpass Filter

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Bandpass Filter

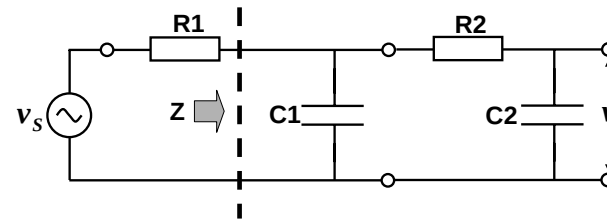
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Bandpass Filter

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Dubbelt RC-nät

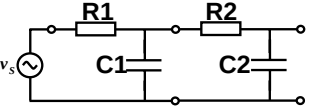
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Dubbelt RC-nät -- Lågpas

$$Z = \left(R_2 + \frac{1}{sC_2} \right) \parallel \frac{1}{sC_1}$$

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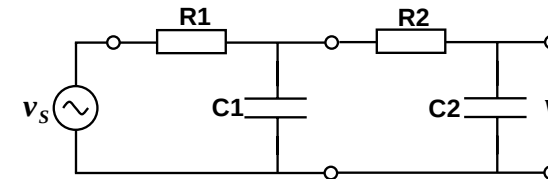
Dubbelt RC-nät -- Lågpas



$$H(s) = \underbrace{\frac{1}{sC1} + R2 + \frac{1}{sC2}}_{\text{Spänningsdelning R1 - resten}} \cdot \underbrace{\frac{1}{sC2}}_{\text{Spänningsdelning C2 - R2}} \cdot \frac{1}{R2 + \frac{1}{sC2}}$$

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Dubbelt RC-nät -- Lågpas

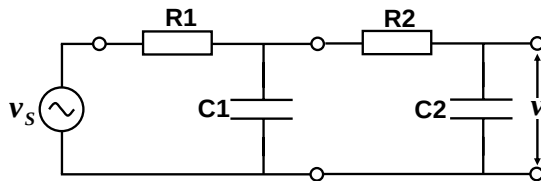


$$H(s) = \frac{1}{s - p_1} \frac{1}{s - p_2}$$

p_1 och p_2 är poler till systemet

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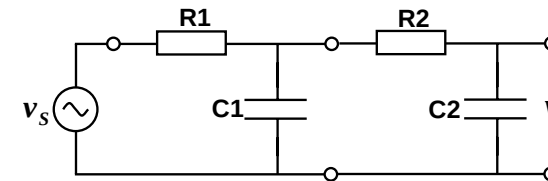
Dubbelt RC-nät -- Lågpas



$$p_1, p_2 = -\frac{R1C1 + R2C2 + R1C2}{2R1C1R2C2} \pm \sqrt{\left(\frac{R1C1 + R2C2 + R1C2}{2R1C1R2C2}\right)^2 - \frac{1}{R1C1R2C2}}$$

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Dubbelt RC-nät -- Lågpas

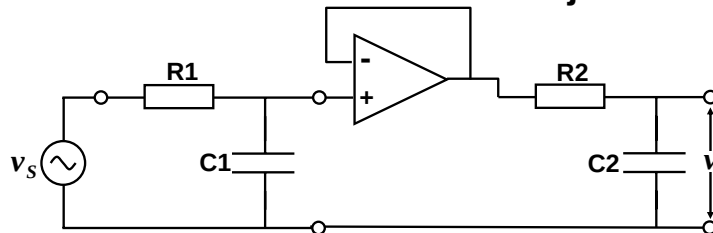


Om $R2 \gg R1$ kan RC-länkarna studeras oberoende med 2 brytpunkter

$$f_1 \approx \frac{1}{2\pi R1C1}, \quad f_2 \approx \frac{1}{2\pi R2C2}$$

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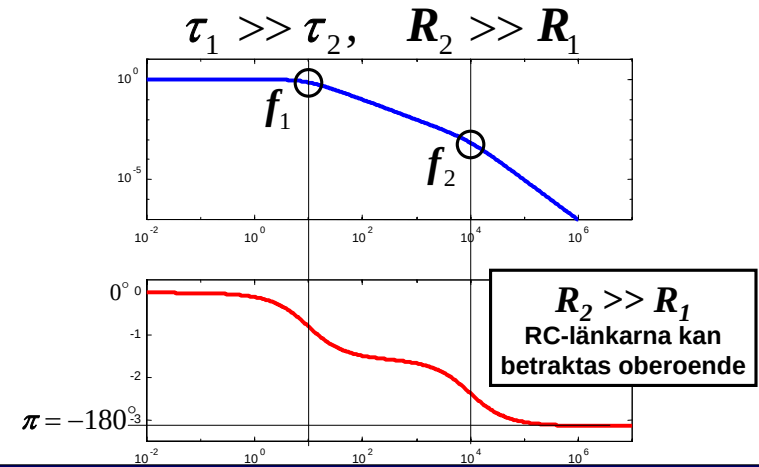
Dubbelt RC-nät med följare



Om en följare separerar första och andra länken $\Rightarrow$
2 brytpunkter oberoende

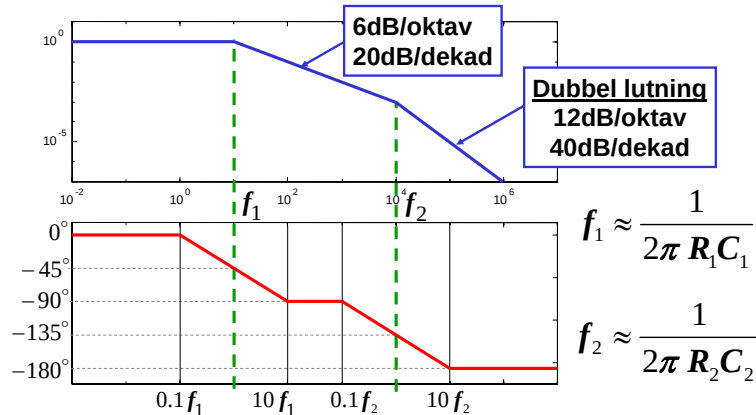
$$f_1 \approx \frac{1}{2\pi R_1 C_1}, \quad f_2 \approx \frac{1}{2\pi R_2 C_2}$$

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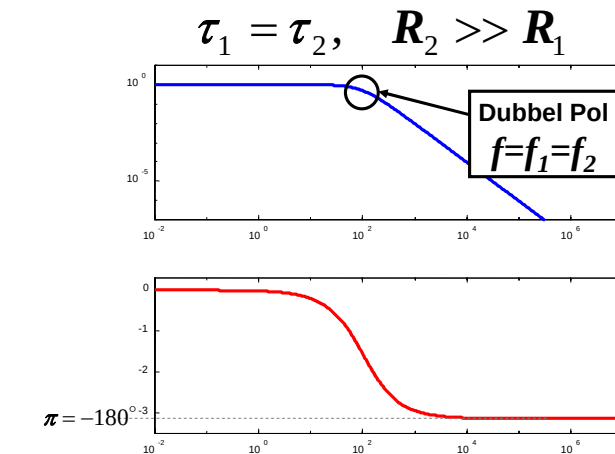


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Asymptotiskt, $\tau_1 \gg \tau_2, \quad R_2 \gg R_1$

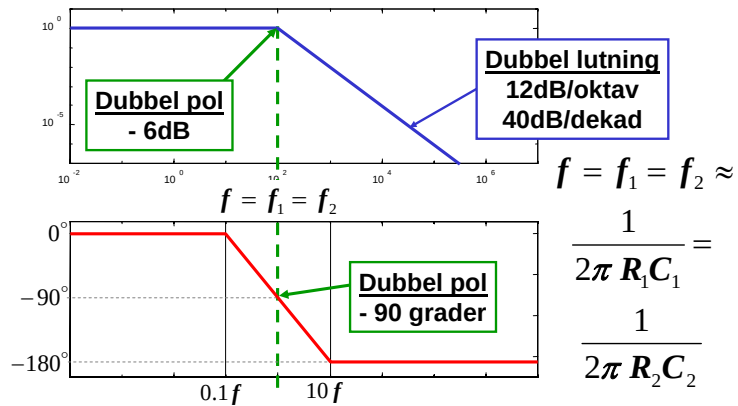


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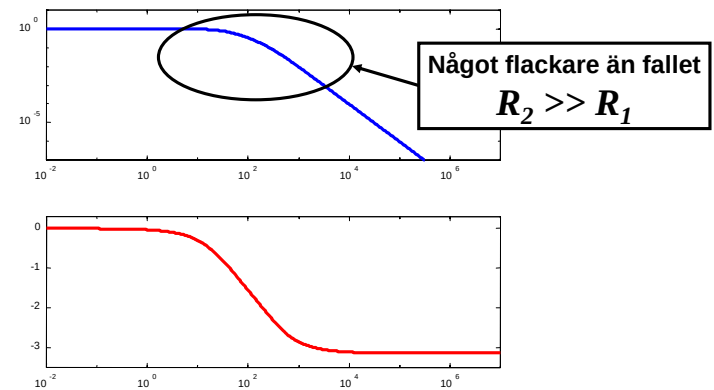
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Asymptotiskt, $\tau_1 = \tau_2$, $R_2 \gg R_1$



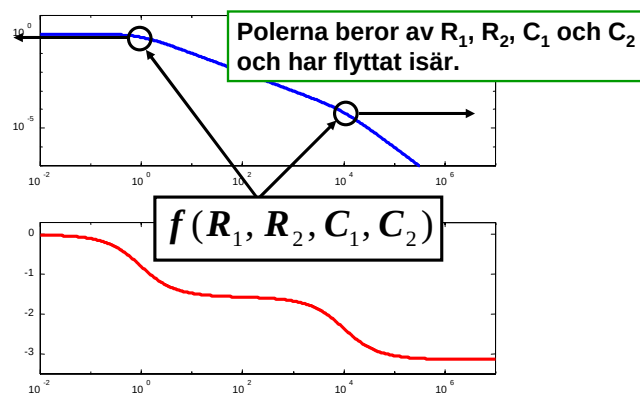
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$\tau_1 = \tau_2$, $R_2 = R_1$



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$\tau_1 = \tau_2$, $R_2 \ll R_1$



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